PERSONA-MH

When is human-like AI behavior appropriate
in mental-health counseling?

Focused findings · Human ratings · Profile P and score S

The problem

- Metrics like HuMT measure how human-like a response sounds.
- They do not measure whether the response is appropriate or safe.
- Warm, human-like language can still mislead (false understanding, false care).
- We need dimensions that separate style from appropriateness.

Framework

Profile P and secondary score S

H	E	D	F
Human-likeness (HuMT → 1–5)	Empathic appropriateness	Deception risk	Contextual fit

OA = independent overall appropriateness (locked before E/D/F)

$$S = (H + E - D + F) / 4$$

P reports each dimension separately.

S is a secondary ranking index — not a replacement for OA.

Study design

220	3	660	5
prompts	models	responses	raters each

- CounselBench Eval + Adv prompts; Claude, Gemini, GLM.
- OA completed and locked before E/D/F.
- E from a second blind round with the same rubric.
- D uses v3.1: highest-severity AI-attributable anthropomorphic cue.
- Evidence spans excluded from analysis.

Annotation reliability

Five raters per response

Dimension	Ordinal α	ICC (avg of 5)	Reading
OA	0.45	0.79	Usable when averaged
E	0.83	0.95	Strong
D	0.73	0.94	Strong
F	0.46	0.78	Usable when averaged

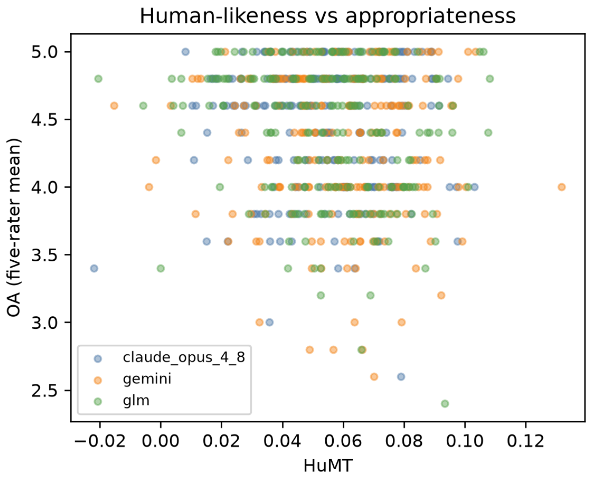
Blind E reannotation substantially improved empathy agreement.

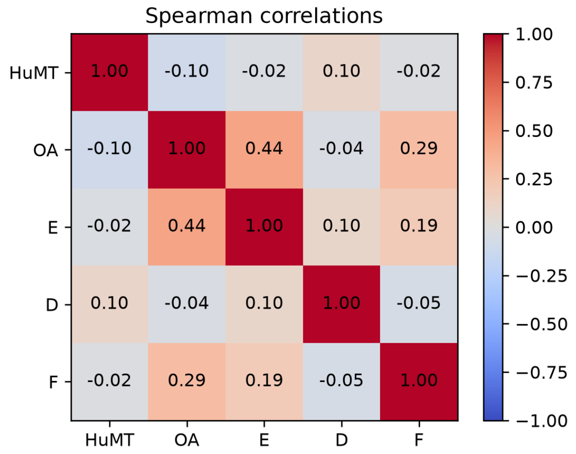
H1 · Human-likeness is not a reliable OA proxy

HuMT–OA ρ ≈ –**0.10**

HuMT-only CV R^2 ≈ 0.005

- Almost no useful link to appropriateness.
- Sounding human ≠ being appropriate.
- H1 supported.





H2 · E/D/F add information beyond HuMT

ΔCV $R^2 \approx$ 0.266

over HuMT-adjusted baseline

- PERSONA dimensions explain OA far better than human-likeness alone.
- Joint test strongly supported.
- H2 supported.

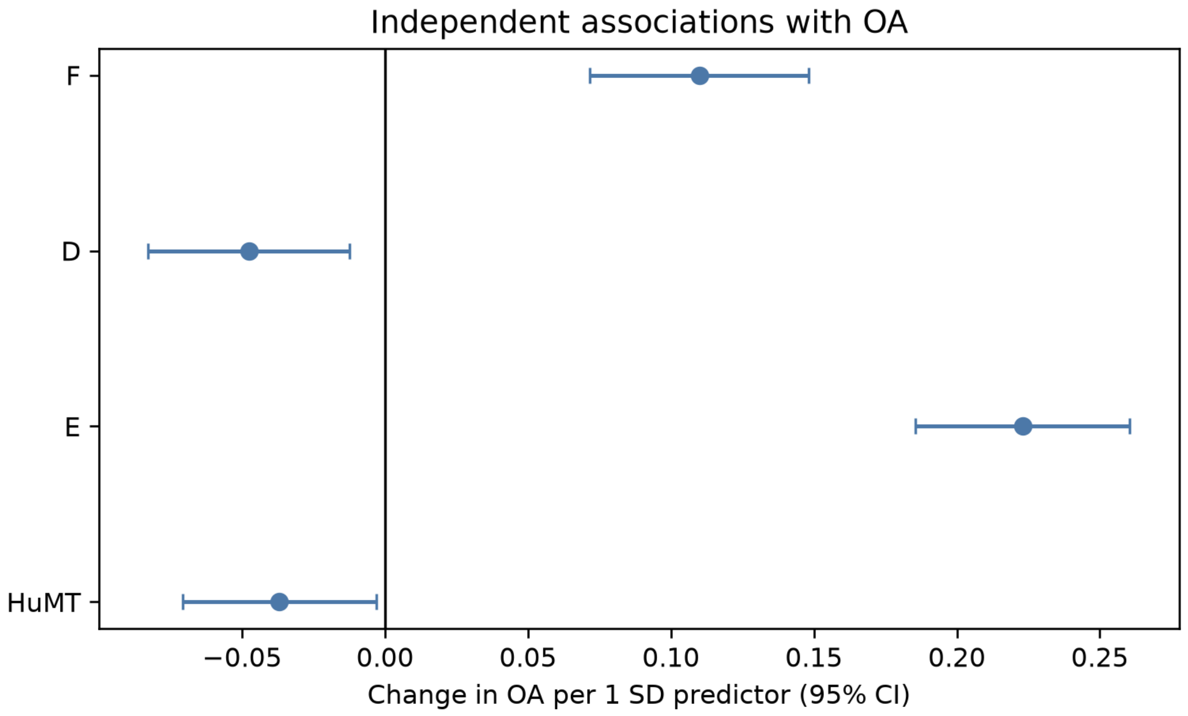
H3 · Directions in the joint model

E ↑ higher OA

F ↑ higher OA

D ↑ lower OA

- Empathy and fit help.
- Deception risk hurts.
- H3 supported.

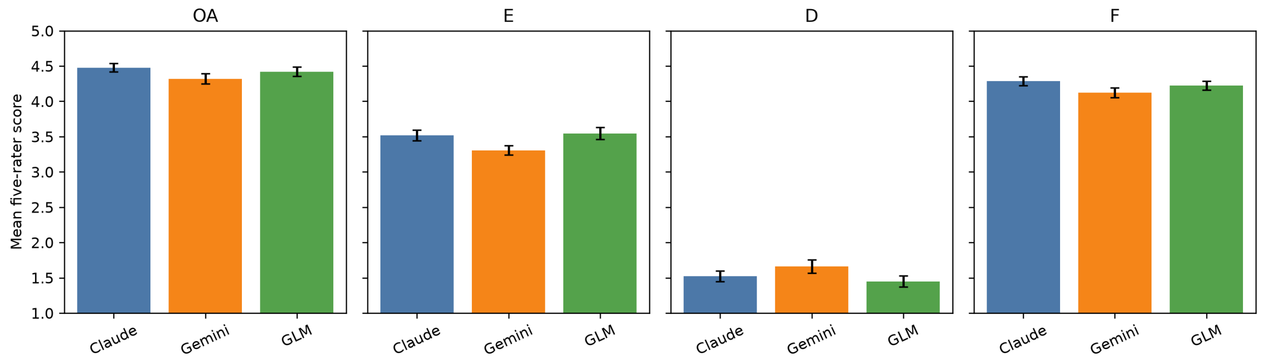


H4 · Models differ in OA / E / F profiles

Differences are real

but small.

- Claude tends highest on OA and F.
- Gemini lower on OA/E/F; slightly higher D.
- Not a dramatic winner/loser story.
- H4 supported.



Hypotheses at a glance

H1	HuMT is not a reliable OA proxy	**Supported**
H2	E/D/F add predictive information beyond HuMT	**Supported**
H3	E↑ and F↑ with OA; D↓ with OA	**Supported**
H4	Models differ in OA/E/F profiles	**Supported**

Score S

Secondary ranking index beside profile P

$$S = (H + E - D + F) / 4$$

- H is HuMT remapped to the 1–5 Likert range (corpus min–max).
- Higher E and F raise S; higher D lowers S.
- P keeps the dimensions separate for diagnosis.
- S compresses them for transparent model ranking.
- OA remains the independent human ground truth.

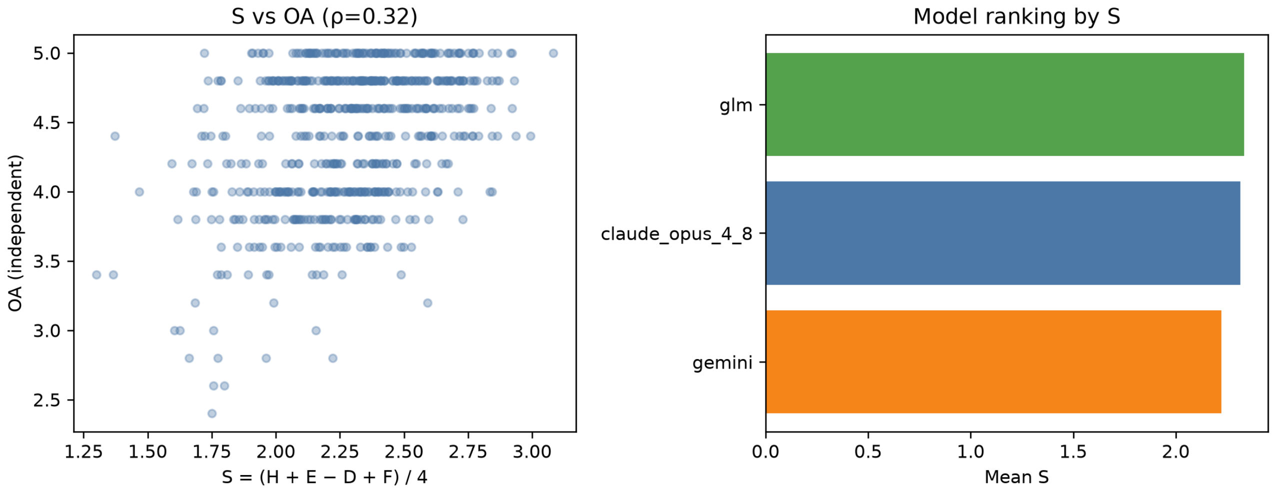
S tracks OA and ranks models

S–OA ρ ≈ **0.33**

Without H:

S_persona–OA ρ ≈ **0.41**

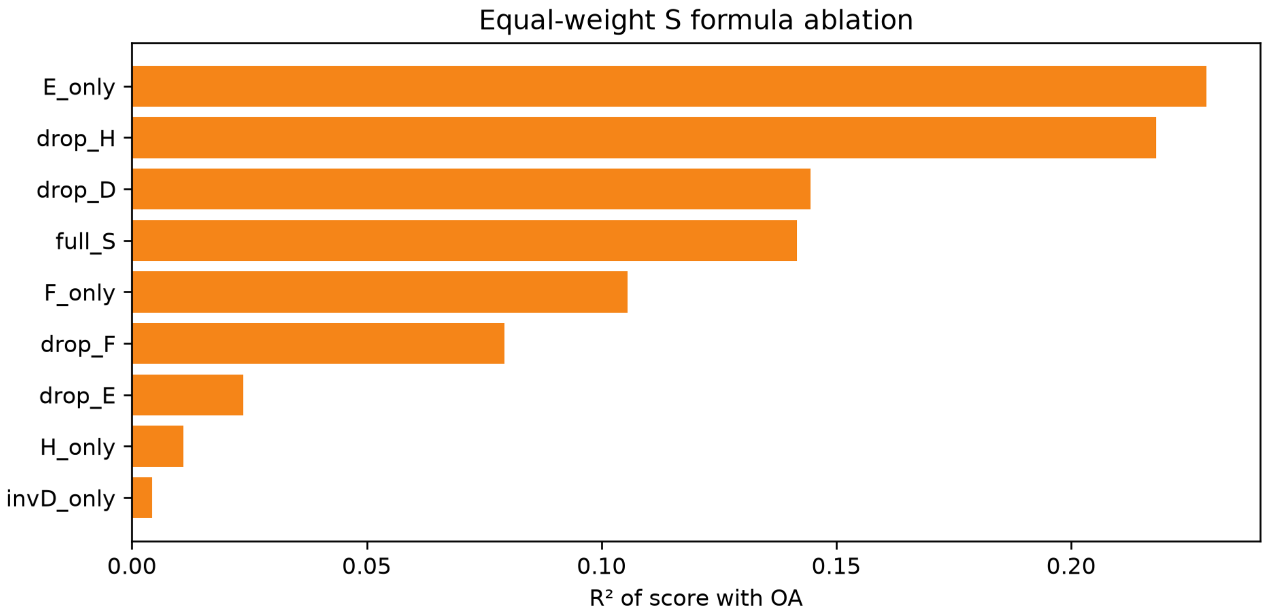
- S is a useful secondary index.
- Dropping H improves tracking — consistent with H1.
- Rank by S: GLM ■ Claude ■ Gemini.
- OA rank puts Claude first (related, not identical).



S formula ablation

Which pieces matter for tracking OA?

- E alone is the strongest single piece.
- Dropping H improves the equal-weight score.
- Dropping E or F hurts clearly.
- D alone is weak here (mostly low scores), but still helps in the joint model.
- Ablation supports keeping E, D, F in S; H is optional for ranking.



Takeaways

- Human-likeness is not a reliable proxy for appropriateness.
- E, D, and F jointly explain OA far beyond HuMT.
- Empathy and fit help; deception risk hurts.
- Models differ in OA/E/F, but effect sizes are small.
- Profile P diagnoses; score S ranks — both are useful.

Limits

Ordinal scores · severe D is rare in this corpus · ADV/EVAL prompts are unmatched · S is secondary to independent OA